

Reading the Liver Panel.

Almost every child you see in hepatology clinic arrives with a liver panel. It was usually ordered for another reason: before surgery, during a febrile illness, as part of an obesity check, or when a new medication was started. The panel is often called liver function tests, but the name is misleading. Most of what it measures is injury, not function. ALT, AST, alkaline phosphatase and GGT are enzymes. They leak into the blood when cells are damaged, or rise when bile flow is blocked. The tests that reflect what the liver actually does are the INR, which reflects the synthesis of clotting factors, albumin, which reflects protein synthesis, and, in part, bilirubin. A child can have an ALT of two thousand and recover completely. Another child can have a normal ALT and be close to liver failure.

In this chapter we will learn a structured way to read a liver panel in a child. It rests on three questions, asked in order. What is the pattern? How big is the abnormality? And is it really abnormal for this particular child? After that, we will go through the common traps, what the evidence says about which abnormalities persist, and how the four major guidelines differ. These are the American College of Gastroenterology guideline from 2017, the British Society of Gastroenterology guideline from 2018, the AASLD paediatric guidance from 2025, and the ESPGHAN position paper from 2026.

Part one. Where the enzymes come from.

Two facts explain almost every puzzling liver panel. The first is where the enzyme sits inside the cell, and in which tissues. The second is how long it stays in the blood, which is its half-life. Learn these before anything else.

ALT lives in the cytoplasm of the liver cell. There is only a little in kidney and muscle. That makes ALT the most liver-specific marker of liver cell injury that we use every day. Its half-life is about forty-seven hours.

AST is found both in the cytoplasm and in the mitochondria, and there is a lot of it outside the liver: in heart, in skeletal muscle, in red blood cells and in brain. Its half-life is much shorter, about seventeen hours. Two practical consequences follow. First, after any acute liver injury, AST falls before ALT. So an AST that is dropping faster than the ALT is expected during recovery. Second, when AST is clearly higher than ALT, think first of a source outside the liver, such as muscle or haemolysis, or of macro-AST, or of advanced liver disease. Only after that should you think of ordinary hepatitis.

Alkaline phosphatase is a membrane enzyme. In the liver it lines the canalicular membrane of the liver cell and the lining of the bile ducts. But alkaline phosphatase also comes from bone, and from intestine, placenta and kidney. In children, bone is

the main source, because the growth plates are active. This single fact explains three things: why alkaline phosphatase is much higher in children than in adults, why it peaks during the growth spurt of puberty, and why a high alkaline phosphatase on its own is so often not a liver problem at all. It can also rise after a fatty meal in people with blood group O or B, because of the intestinal form of the enzyme.

GGT sits on the canalicular membrane and in the cells lining the bile ducts. It is also found in kidney and pancreas, but not in bone. That makes GGT the test that tells you whether a high alkaline phosphatase comes from the liver. If both are raised, the source is the liver and bile ducts. If alkaline phosphatase is high and GGT is normal, look at bone. GGT is also raised by some drugs, such as anticonvulsants. In an infant with cholestasis, a low or normal GGT is itself a diagnostic clue. It points to a small group of disorders in which bile flow fails without damage to the bile ducts: progressive familial intrahepatic cholestasis type one and type two, and bile acid synthesis defects.

Two more enzymes can help. GLDH, glutamate dehydrogenase, is a mitochondrial enzyme found almost only in the liver, mainly around the central vein. It is not available everywhere, but where it is, it answers a useful question. In a boy with muscular dystrophy, ALT can be twenty times normal while GLDH stays normal. LDH is present in every cell of the body. Its main use is a ratio. If the ALT divided by the LDH is less than one and a half, think of ischaemic injury, the so-called shock liver, or of paracetamol poisoning, rather than viral hepatitis.

Part two. What is the pattern?

The first question is whether the injury is mainly to the liver cells, which we call hepatocellular, or to bile flow, which we call cholestatic. A number called the R ratio helps. It was first designed for drug-induced liver injury. To calculate it, take the ALT as a multiple of its upper limit of normal, and divide it by the alkaline phosphatase as a multiple of its upper limit of normal.

If R is five or more, the pattern is hepatocellular. ALT dominates, and the typical causes are viral hepatitis, autoimmune hepatitis, Wilson disease, drug injury and ischaemia. In the large paediatric drug-induced liver injury cohort, seventy-eight percent of cases were hepatocellular. If R is between two and five, the pattern is mixed. Both enzymes are raised, and you should think of infiltration of the liver, drug injury and early obstruction. That was nine percent of cases. If R is two or less, the pattern is cholestatic. Alkaline phosphatase and GGT dominate, and you should think of biliary obstruction, sclerosing cholangitis and drugs. That was thirteen percent of cases.

The R ratio is only as good as the number you divide by. You must use the alkaline phosphatase limit for the child's age and sex. If you use the adult limit, which is about

one hundred and twenty-six units per litre, almost every growing child will look cholestatic.

Here is an example. A fourteen-year-old boy has an ALT of ninety, with an upper limit of thirty-five, and an alkaline phosphatase of four hundred. If you use the adult limit of one hundred and twenty-six, his alkaline phosphatase looks three times the upper limit, and his R comes out below one. That looks cholestatic. But the real limit for a boy of his age is about five hundred and twenty-five. Against that limit his alkaline phosphatase is normal, and his R is about three and a half. The adult limit made his R about four times too small, and the wrong pattern would send the whole work-up in the wrong direction.

Part three. How big is the abnormality?

The American guideline grades a raised ALT or AST in five bands, and each band has its own short list of causes.

A borderline rise is less than twice the upper limit of normal. In children, the commonest causes are fatty liver disease, coeliac disease, muscle, drugs, thyroid disease and recent exercise. Useful tests to add are creatine kinase, coeliac serology, thyroid stimulating hormone and a careful drug history.

A mild rise is two to five times the upper limit. Add autoimmune hepatitis, Wilson disease, alpha-1 antitrypsin deficiency, and chronic hepatitis B or C. Useful tests are immunoglobulin G, autoantibodies, ceruloplasmin and alpha-1 antitrypsin.

A moderate rise is five to fifteen times the upper limit. Think of acute viral hepatitis, meaning hepatitis A, hepatitis E, Epstein-Barr virus and cytomegalovirus, and also of autoimmune hepatitis and drug injury. Add acute viral serology and an INR.

A severe rise is more than fifteen times the upper limit. Think of paracetamol, ischaemia from heart failure or sepsis, herpes simplex virus, and autoimmune hepatitis presenting as acute liver failure. Add a paracetamol level, LDH, an echocardiogram and a herpes simplex PCR.

A massive rise, above ten thousand units per litre, means paracetamol or ischaemic hepatitis. But also think of rhabdomyolysis and heat stroke, and measure creatine kinase, because the liver may not be the source at all.

There is an important warning, which the British guideline states clearly. The size of the rise tells you about the cause, not about the prognosis. Hepatitis A can push the ALT above one thousand and resolve without a trace. Hepatitis C can progress to cirrhosis with an ALT that is always normal. A small number does not mean a small disease.

Part four. Is it really abnormal for this child?

A result is only abnormal compared with a reference range, and in children that range moves with age and sex. For ALT, most laboratory ranges are also simply too high.

Laboratory reference ranges are built from people who appear healthy. But in a population where fatty liver disease is common, that group already contains many children with liver disease, and the upper limit is pulled upwards. The SAFETY study, published by Schwimmer in 2010, measured the size of the problem. The median upper limit of ALT used by American children's hospitals was fifty-three units per litre, with a range from thirty to ninety. In healthy-weight children without liver disease, the ninety-fifth centile was about twenty-six in boys and twenty-two in girls. With the hospital limits, only about a third to a half of children with fatty liver disease, hepatitis B or hepatitis C were detected. With the lower, biology-based limits, about three quarters were detected, and specificity fell only a little. The lesson for daily practice is this: an ALT inside the laboratory's normal range, in a child with obesity, does not mean that the liver is normal.

Unfortunately, the guidelines do not agree on a single number. The American adult guideline uses twenty-nine to thirty-three for men and nineteen to twenty-five for women. The British guideline accepts the local laboratory range, while admitting it is probably too high. The AASLD paediatric guidance uses twenty-six for boys and twenty-two for girls from the age of ten, and thirty for younger children. ESPGHAN uses a single limit of thirty for all ages and both sexes. A practical approach is to read the ALT against twenty-six for boys and twenty-two for girls, whatever the laboratory prints.

GGT changes dramatically in the first years of life. It is very high at birth and falls steeply during infancy. At one American paediatric laboratory, the upper limit is one hundred and sixty-eight in the first month of life, one hundred and twenty-six at one to six months, forty-two in the second half of infancy, and only sixteen between one and six years. After that it rises slowly towards adult values. So a GGT of one hundred and twenty is normal in a three-week-old baby, but more than seven times the upper limit in a three-year-old.

This has a direct clinical use. In a jaundiced infant with conjugated hyperbilirubinaemia, the GGT is expected to be high. A GGT that is normal for age in a cholestatic infant is the abnormal finding, and it should make you think of progressive familial intrahepatic cholestasis and bile acid synthesis defects.

Alkaline phosphatase follows the growth plate. Its upper limit is about three hundred to four hundred and twenty through early childhood, and it peaks at about five hundred and thirty to five hundred and sixty at ten to eleven years. After that, boys and girls diverge. Girls enter puberty earlier and fall towards adult values by about

sixteen. Boys stay above five hundred until about fifteen. So the same alkaline phosphatase of four hundred is normal in a fourteen-year-old boy, but almost twice the upper limit in a fourteen-year-old girl.

When alkaline phosphatase is raised on its own, the order of work-up is simple. Check GGT first, to ask whether it is the liver. Then check vitamin D, to ask whether it is rickets. And consider growth and transient hyperphosphatasemia before going any further.

Part five. The common traps.

The first trap is an isolated high AST in a well child. Sometimes AST binds to an immunoglobulin and forms a large complex called macro-AST. The body cannot clear it, so the AST stays high, while ALT, creatine kinase and LDH are normal, and the child is completely well. It is harmless, but it is often investigated extensively, sometimes as far as a liver biopsy. In one series of forty-four well children with an isolated high AST, seventeen, nearly forty percent, had macro-AST. The screening test is precipitation with polyethylene glycol, and electrophoresis confirms the diagnosis. Most children remain positive for years, and all stay well. Macro-AST is often familial and genetic, so a cheap first step is to measure the parents' AST before investigating the child's liver.

The second trap is muscle. ALT is liver-specific enough for most purposes, but not when a large amount of muscle is breaking down. In boys with Duchenne muscular dystrophy who were still walking, ALT reached more than twelve hundred and AST eight hundred. In one report, six boys had their diagnosis delayed because they underwent a liver work-up before anyone measured creatine kinase. In a large Italian cohort of children with persistently raised aminotransferases, fourteen of the fifty-one genetic diagnoses were muscular dystrophies. Exercise matters too. After one hour of weightlifting, ALT and AST can stay raised for a week or more. So, in any boy with a raised ALT and AST, measure creatine kinase. It is the cheapest test in the panel, and it can save a child from an unnecessary liver work-up.

The third trap is a very high alkaline phosphatase in a well toddler. This is usually transient hyperphosphatasemia of infancy and early childhood, and it is benign. In a review of more than eight hundred published cases, about seventy percent had an alkaline phosphatase at least five times the upper limit, sometimes in the thousands, and eighty percent returned to normal within four months. It affects up to three and a half percent of children between two months and two years of age, and it often follows an infection. The Kraut criteria define it: age under five, no evidence of bone or liver disease, and a return to normal within four months. The right response is to check that GGT, ALT, calcium, phosphate and vitamin D are normal, and then simply repeat the alkaline phosphatase in three to four months.

The fourth trap is that low values can be abnormal too. We are trained to scan the panel for high numbers, but some of the most important findings are low ones.

A low alkaline phosphatase for age is the clue to hypophosphatasia, a genetic disorder of bone mineralisation. Children are often diagnosed more than a year after the first sign. Look for early loss of baby teeth, skeletal deformity and poor growth.

Alkaline phosphatase needs zinc to work, so zinc deficiency lowers it. This happens in malnutrition, intestinal failure and chronic cholestasis. In a cholestatic child with a normal alkaline phosphatase, check zinc before accepting the value.

In Wilson disease presenting as acute liver failure, a low alkaline phosphatase alongside haemolysis and a very high bilirubin is a classic clue. A normal or low alkaline phosphatase in a jaundiced child is a red flag.

And a falling ALT in a sick child is not always good news. When ALT falls while the INR and bilirubin rise, the liver has lost its mass of liver cells. There are simply no cells left to release the enzyme. This is deterioration, not recovery.

The rule to remember is this: an alkaline phosphatase that is too low for the child's age should never be reported as normal.

Two ratios have been proposed to recognise Wilson disease in acute liver failure. The first is alkaline phosphatase divided by bilirubin in milligrams per decilitre, below four. The second is AST divided by ALT, above two point two. In the original American cohort, mostly adults, each ratio detected almost all cases. But in a study of forty children from Bangladesh, the first ratio detected only about a third of cases, and the second about three quarters. Specificity stayed high. So, in a child, a positive ratio supports Wilson disease, but a negative ratio does not exclude it. Whenever Wilson disease is possible, send ceruloplasmin and urinary copper, and ask for a slit-lamp examination of the eyes.

Part six. What persists, and what is found.

Two studies give a realistic picture of what a raised ALT means in children. They also show how much depends on where the child is seen.

In Naples, Iorio and colleagues followed four hundred and twenty-five children referred to a specialist centre with isolated raised aminotransferases. Sixty-one percent returned to normal within six months. Among the one hundred and sixty-six who did not, the commonest cause was obesity-related liver disease, in seventy-five children. Fifty-one children had a genetic disorder. These included Wilson disease in eighteen, muscular dystrophy in fourteen, and, less often, alpha-1 antitrypsin deficiency, Alagille syndrome, hereditary fructose intolerance and glycogen storage disease. A smaller number had gallstones or a choledochal cyst, autoimmune

hepatitis or coeliac disease, and in twenty-two no cause was found. So about one in three children with a persistent abnormality had a genetic disorder, and many of these, like Wilson disease, are treatable. The message is: repeat the test first, but take the second abnormal result seriously. That is where the treatable diseases are found.

In a different setting, Rudolph and colleagues studied one hundred and twenty obese children without symptoms, each with an ALT of forty or more, who all had a full screening panel. Only one child had a diagnosis other than fatty liver disease, and that was muscular dystrophy. None had Wilson disease, autoimmune hepatitis, viral hepatitis, coeliac disease or alpha-1 antitrypsin deficiency. None of seventy ultrasound scans showed a significant abnormality. The main result of broad testing was false positives.

Put together, these two studies show that the likelihood of disease before you test decides the panel. In a specialist cohort, persistent abnormalities often have a genetic cause. In an obese child in primary care, the one test that reliably pays off is creatine kinase.

Part seven. What the guidelines say.

The first question is whether to repeat the test or investigate straight away. Here the adult guidelines openly disagree. The American guideline says to repeat the panel, or add a clarifying test such as GGT, before starting a full evaluation. The British guideline says to investigate an abnormal result regardless of its level or duration, because most abnormal results in primary care were still abnormal a month later, and even two years later. For children, the British guideline advises a low threshold for referral to a paediatric hepatologist.

The paediatric guidelines take a middle path. The AASLD says that in a well child with a single raised ALT, you should repeat it within three months, but if acute hepatitis is suspected, repeat it within one to two days and refer. ESPGHAN says that an ALT between thirty and eighty should be repeated in two to six months, an ALT of eighty or more should be investigated now, and any red flag, such as jaundice, a prolonged INR, confusion, a large liver or spleen, or poor growth, means immediate referral. The practical rule for children is: repeat, but not twice. The second abnormal result starts the search for a cause.

The guidelines also differ on what to include in that search. Combining them, a sensible first-line screen for a child with a persistently raised ALT includes GGT; creatine kinase; coeliac serology with total IgA; immunoglobulin G with autoantibodies, including antinuclear antibody, anti-smooth muscle antibody and anti-LKM antibody; ceruloplasmin from the age of three; alpha-1 antitrypsin level or phenotype; thyroid function; hepatitis B and hepatitis C serology; INR and albumin;

and an abdominal ultrasound. Iron studies add little in children. And the panel should be adjusted to the child in front of you. In an obese adolescent with a borderline ALT, the list can be much shorter.

The guidelines agree on some points. All use GGT to confirm that a raised alkaline phosphatase comes from the liver. All suggest thinking of muscle when AST is disproportionately high. And all accept that fibrosis scores developed in adults have not been properly validated in children. Only the British guideline specifically names low-GGT cholestasis, and only the paediatric documents properly address age and sex differences in normal values.

Part eight. Reading the panel like an expert.

Experienced hepatologists read the liver panel as a series of results over time, not as a single snapshot. Four observations are especially useful.

The first is kinetics. Because AST is cleared faster than ALT, AST falls first after an acute injury. Conjugated bilirubin can also bind permanently to albumin, forming what is called delta bilirubin. It shares albumin's half-life of about three weeks, which is why jaundice can linger for weeks after the liver has recovered.

The second is LDH. If the ALT to LDH ratio is below one and a half, think of ischaemic or paracetamol injury rather than viral hepatitis. Check it in any child with an ALT in the thousands.

The third is the INR. In cholestasis, the INR can be prolonged because vitamin K, a fat-soluble vitamin, is not absorbed, not because the liver cannot make clotting factors. So give intravenous vitamin K and repeat the INR before you call it liver failure. An INR that does not correct means true failure of liver synthesis.

The fourth is albumin. Albumin also has a half-life of about three weeks, so a low albumin means an illness lasting weeks, or losses through the gut or the kidney. It also falls in any severe illness because of inflammation, so it is a poor marker of acute liver failure.

Bilirubin itself, with its own physiology and its own paediatric emergencies such as neonatal cholestasis, is the subject of the next chapter.

Part nine. A practical approach.

Let us put it all together as a sequence of steps.

First, ask whether the child is sick. Jaundice, confusion, bleeding, an INR above one and a half, or an ALT more than fifteen times normal is not a reading exercise.

Assess the child urgently for acute liver failure.

Second, use the right normal values. Read the ALT against twenty-six for boys and twenty-two for girls, and read alkaline phosphatase and GGT against the limits for age and sex.

Third, name the pattern, using the R ratio with paediatric limits: hepatocellular, mixed or cholestatic.

Fourth, grade the size of the rise, and use it to narrow the list of causes.

Fifth, look for the traps. If AST is higher than ALT, check creatine kinase and think of macro-AST. If alkaline phosphatase is raised alone, check GGT and vitamin D, and think of growth and transient hyperphosphatasemia. If alkaline phosphatase is low, think of hypophosphatasia, zinc deficiency and Wilson disease.

Sixth, check liver function: the INR, after vitamin K if the child is cholestatic, the albumin and the bilirubin.

Seventh, in a well child with an isolated mild rise, repeat the test once, within a few weeks to three months. If it is still abnormal, start the search for a cause.

And eighth, read the trend. A falling ALT is good news only if the INR and bilirubin are falling too.

Part ten. Four clinical cases.

Case one. A four-year-old boy has blood tests before a tonsillectomy. His ALT is one hundred and eighty and his AST two hundred and twenty. He is well, but his mother says he falls more than other children and struggles to climb stairs. A hepatitis screen is being arranged. What would you do? Measure creatine kinase before anything else. It comes back at fourteen thousand. The aminotransferases are coming from muscle. The next step is referral to a neuromuscular specialist for suspected Duchenne muscular dystrophy, not a liver work-up.

Case two. A fifteen-month-old girl had gastroenteritis two weeks ago. A blood test during that illness showed an alkaline phosphatase of two thousand four hundred. Her ALT, GGT, bilirubin, calcium, phosphate and vitamin D are all normal, she is growing well, and her examination is normal. What would you do? She fits the criteria for transient hyperphosphatasemia. Reassure the family and repeat the alkaline phosphatase in three to four months. No imaging or further tests are needed if the repeat is normal.

Case three. A thirteen-year-old girl presents with jaundice and tiredness. Her bilirubin is twenty milligrams per decilitre, AST one hundred and eighty, ALT seventy, alkaline phosphatase only forty, haemoglobin eight and a half with a negative Coombs test, and INR two point four. What is the pattern? The alkaline phosphatase is far too low for her age. The alkaline phosphatase to bilirubin ratio is two, which is

below four. The AST to ALT ratio is more than two and a half. And there is haemolysis that is not immune. This is Wilson disease presenting as acute liver failure until proven otherwise. She needs urgent referral to a transplant centre, together with ceruloplasmin, urinary copper and a slit-lamp examination.

Case four. A fourteen-year-old boy with obesity has an ALT of thirty-eight. The laboratory reports it as normal, because its limit is forty-five. Is it normal? No. Against a biology-based limit of twenty-six for boys, his ALT is about one and a half times the upper limit. In an obese adolescent, this is most likely fatty liver disease. It should be recorded as abnormal and repeated. If it persists, he should have a focused screen, including creatine kinase, coeliac serology and thyroid function, and be managed according to paediatric fatty liver guidance.

Key points to remember.

Most of the so-called liver function tests measure injury, not function. Function is shown by the INR, albumin and bilirubin.

ALT is specific to the liver. AST also comes from muscle, heart and red cells. Alkaline phosphatase also comes from bone. GGT does not come from bone, and it confirms that a high alkaline phosphatase comes from the liver.

Use the R ratio to name the pattern, and always use paediatric limits for alkaline phosphatase.

The size of the rise narrows the list of causes, but it does not predict the outcome.

Healthy children have an ALT below about twenty-six in boys and twenty-two in girls. Many laboratory limits are too high.

GGT is very high in newborns and low in toddlers. Alkaline phosphatase is high throughout childhood and peaks in early puberty.

Measure creatine kinase in children with raised aminotransferases, especially boys. Think of macro-AST when AST is raised alone, and of transient hyperphosphatasaemia when a well toddler has a very high alkaline phosphatase on its own.

A low alkaline phosphatase, a normal GGT in a cholestatic infant, and a falling ALT with a rising INR are all dangerous findings.

Repeat an isolated mild abnormality once. If it persists, investigate systematically.

Questions to test yourself.

Question one. Why does AST fall faster than ALT after an acute liver injury? Answer: because its half-life is about seventeen hours, compared with about forty-seven hours for ALT.

Question two. A three-week-old baby has a GGT of one hundred and twenty. Is this abnormal? And what about a three-year-old with the same value? Answer: in the three-week-old it is normal. In the three-year-old it is more than seven times the upper limit.

Question three. A jaundiced two-month-old has conjugated hyperbilirubinaemia and a normal GGT. What does this suggest? Answer: low-GGT cholestasis, meaning progressive familial intrahepatic cholestasis types one and two, and bile acid synthesis defects.

Question four. Why can the R ratio mislead in a teenager, and how do you avoid it? Answer: the adult alkaline phosphatase limit makes the alkaline phosphatase look much higher than it really is, so the pattern looks cholestatic. Always use the limit for age and sex.

Question five. A well eight-year-old has an isolated AST of ninety-five on three occasions, with normal ALT and creatine kinase. What is the most likely explanation, and what simple test can you do first? Answer: macro-AST. Measure the parents' AST first, then confirm with polyethylene glycol precipitation.

Question six. A child in intensive care has an ALT that has fallen from three thousand to four hundred over three days, while the INR has risen from one point eight to three point five. How do you interpret this? Answer: this is loss of functioning liver cells, which means worsening acute liver failure, not recovery.

End of chapter.